

GHAZI University DGK

Department Of Computer Science

Course	Programing Fundamentals
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Section	A
Lab No.	03 & 04

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Task # 01

```
# include <iostream>
using namespace std;
int main() {
   int age;
   float pay;
   Char section;
Cout << " Enter age , pay and section";
   cin >> age >> pay >> section;
   Cout<< "Age=" <<age<<endl;
Cout << "Pay = " << pay << endl;
Cout << " Section = " << section << endl;
   return 0;
}
```

Output

Enter age , pay and section
18, 4000, A
Age = 18
pay = 4000
Section = A

Task#2

```
# include <iostream>
using namespace std;
  int main (){

cout << "20/4-2 =" << 20/4-2 << endl;
cout << "6+122-8= " << 6+122-8 << endl;
cout << "4+8*2 =" << 4+8*2 << endl;
cout << "6+17%3-2=" << 6+17%3-2 << endl;
cout << "2+22*(9-7)=" << 2+22*(9-7) << endl;
cout << "(8+7)*2 =" << (8+7)*2 << endl;
cout << "(16+7)%21=" << (16+7)%21 << endl;
cout << "12/(10-6) =" << 12/(10-6) << endl;
cout << "(193)*(2+2)/4=" << (193)*(2+2)/4
                          << endl;

  return 0;
}
```

output::

```
20/4-2=5
6+122-8=120
4+8*2=20
6+17%3-2=6
2+22*(9-7)=46
(8+7)*2=30
(16+7)%21=2
12/(10-6)=3
```

```
(193)*(2+2)/4
=193
```

#3

```
#include <iostream>
using namespace std;
int main() {
int units=5;
float mass =2.5;
double weight;
weight = mass * units;

cout<< "units =" << units << endl;
cout<< "Mass=" << mass<<endl;
cout << "Weight" << weight <<endl;
return 0;
}
```

output

units = 5

Mass = 2.5

Weight = 12.5

Task#4

```
#include <iostream>
using namespace std;
int main (){
int a, b=2;
float c= 4.2;
a= b*c;
cout<<" value of a =" <<a<<endl;
return 0;
}
```

output

value of a= 8

Task#5

```
# include <iostream>
using namespace std;
 int main() {
int qty=10;
 int salesReps=3;
  double unitsEach;
 unitEach = static_cast<double>(qty) /
      salesReps;
Cout << "unitsEach=" << unitsEach << endl;
      return 0;
return 0;
```

output:

unitsEach=3.3333

Task 6

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main(){
srand(time(0));
int num1 = rand() % 900 + 100;
int num2 = rand() % 900 + 100;
cout << "Math Tutor" << endl;
cout << "Solve this addition problem" << endl;
cout << num1 << endl;
cout << "+" << num2 << endl;
cout << "\n Press Enter to see the answer:";
cout << "\n Correct Answer " << endl;
cout << num1 << endl;
cout << "+" << num2 << endl;
cout << "....." << endl;
cout << num1 + num2 << endl;
return 0;
}
```

pul

Math Tutor

Solve this addition problem:

896
+111

Press Enter to see the answer 1007

Correct answer:

896
+111
1007

Hometask

Task 1

Program 1

```
# include <iostream>
using namespace std;
 int main ()
{
 double number1, number2, sum;
Cout << "Enter a number: ";
 Cin >> number1;
Cout << " Enter another number: ";
 Cin >> number2;
 Sum = number1 + number2;
Cout << " The sum of two numbers is "
        << sum;
return 0;
}
```

put

Enter a number: 30

Enter another number: 61

The sum of two numbers is 92

Program 2

```
#include <iostream>
using namespace std;
int main(){
  int number1, number2;
  float quotient;
  cout << "Enter two number and I will
    divide \n";
  cout << "the first by the second For
        you. \n";
  cin >> number1 >> number2;
  quotient = static_cast<float> (number1)/
    number2;
```

```
cout << "Quotient=" << quotient << endl;

    return 0;
}}
```

output

Enter two number and I will
divide the first by the second
for you.

82
33

Quotient = 2.48485.



Q#2

```
#include <iostream>
using namespace std;
int main() {
    int number1 = 20;
    int number2 = 32;
    int number3 = 37;
    int number4 = 24;
    int number5 = 33;
  int sum;
  float average;
  sum = num1 + num2 + num3 + num4 +
        num5;
    average = sum / 5.0;
  cout << "Sum = " << sum << endl;
  cout << "Average = " << average
       << endl;
    return 0;
}
```

output

Sum = 154

Average = 30.8.

```cpp

#include <iostream>
using namespace std;

int main() {
    int age;
    float pay;
    char section;

    cout << "Enter age, pay and section: ";
    cin >> age >> pay >> section;

    cout << "Age = " << age << endl;
    cout << "Pay = " << pay << endl;
    cout << "Section = " << section << endl;

    return 0;
}

```

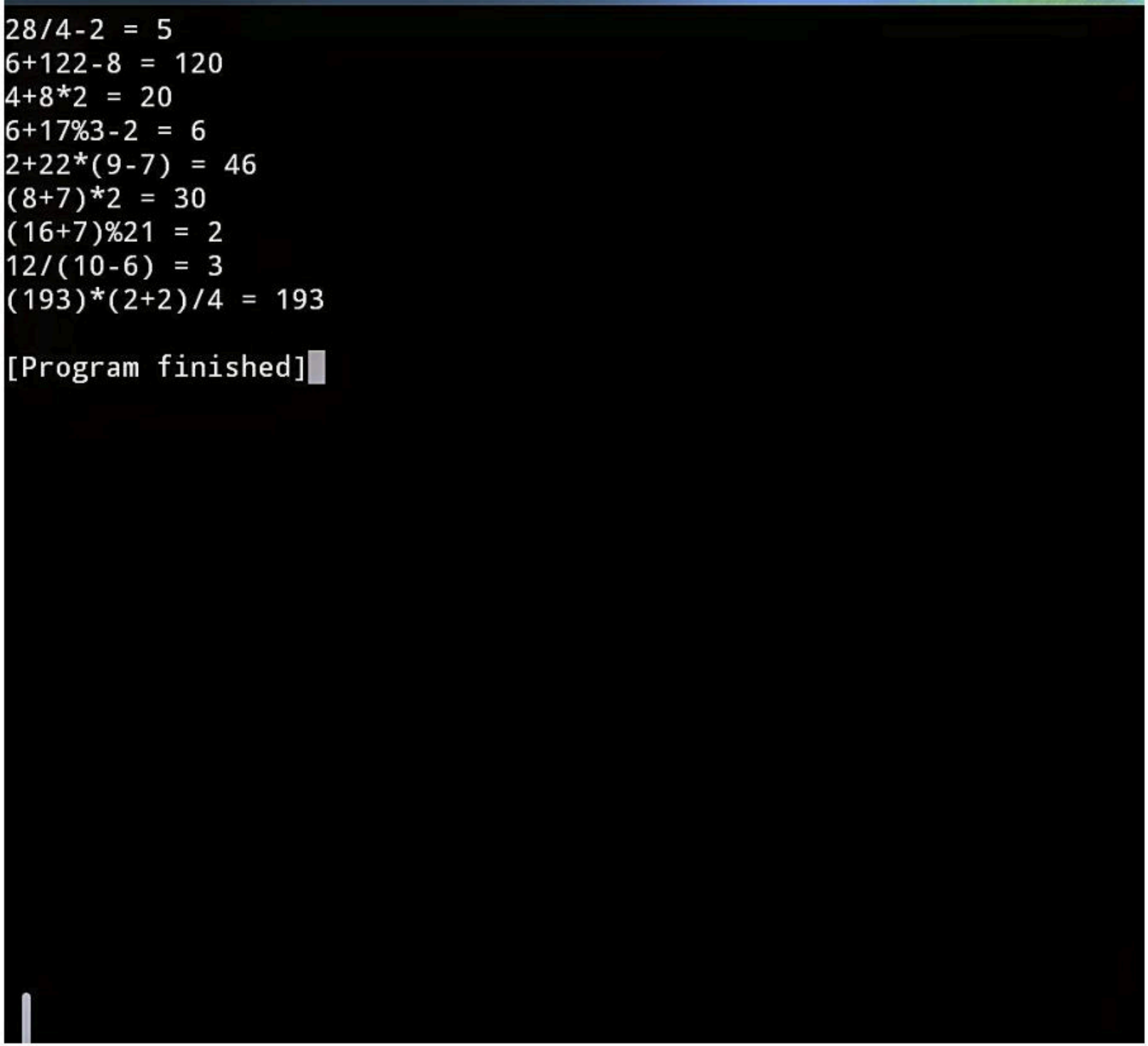
```
Enter age, pay and section: 18 4000 A
Age = 18
Pay = 4000
Section = A

[Program finished]
```

```cpp
#include <iostream>
using namespace std;

int main()
{
    cout << "28/4-2 = " << 28/4-2 << endl;
    cout << "6+122-8 = " << 6+122-8 << endl;
    cout << "4+8*2 = " << 4+8*2 << endl;
    cout << "6+17%3-2 = " << 6+17%3-2 << endl;
    cout << "2+22*(9-7) = " << 2+22*(9-7) << endl;
    cout << "(8+7)*2 = " << (8+7)*2 << endl;
    cout << "(16+7)%21 = " << (16+7)%21 << endl;
    cout << "12/(10-6) = " << 12/(10-6) << endl;
    cout << "(193)*(2+2)/4 = " << (193)*(2+2)/4
<< endl;

    return 0;
}
```


28/4-2 = 5
6+122-8 = 120
4+8*2 = 20
6+17%3-2 = 6
2+22*(9-7) = 46
(8+7)*2 = 30
(16+7)%21 = 2
12/(10-6) = 3
(193)*(2+2)/4 = 193
[Program finished]

```cpp
#include <iostream>
using namespace std;

int main()
{
    int units = 5;
    float mass = 2.5;
    double weight;

    weight = mass * units;

    cout << "Units = " << units << endl;
    cout << "Mass = " << mass << endl;
    cout << "Weight = " << weight << endl;

    return 0;
}
```

```
Units = 5
Mass = 2.5
Weight = 12.5

[Program finished]
```

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a, b = 2;
    float c = 4.2;

    a = b * c;

    cout << "Value of a = " << a << endl;

    return 0;
}
```

```
Value of a = 8

[Program finished]
```

```cpp

#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main()
{
    srand(time(0));

    int num1 = rand() % 900 + 100;
    int num2 = rand() % 900 + 100;

    cout << "Math Tutor" << endl;
    cout << "Solve this addition problem:" << endl;


    cout << num1 << endl;
    cout << "+ " << num2 << endl;

    cout << "\nPress Enter to see the answer...";
    cin.get();

    cout << "\nCorrect Answer:" << endl;
    cout << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "------" << endl;
    cout << num1 + num2 << endl;

    return 0;
}
```

```
Math Tutor
Solve this addition problem:
896
+ 111

Press Enter to see the answer...1005

Correct Answer:
896
+ 111
------
1007

[Program finished]
```

```cpp

#include <iostream>
using namespace std;

int main()
{
    double number1, number2, sum;

    cout << "Enter a number: ";
    cin >> number1;

    cout << "Enter another number: ";
    cin >> number2;

    sum = number1 + number2;

    cout << "The sum of the two numbers is " <<
sum;

    return 0;
}

```

```
Enter a number: 30
Enter another number: 62
The sum of the two numbers is 92
[Program finished]
```

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number1, number2;
    float quotient;

    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";

    cin >> number1 >> number2;

    quotient = static_cast<float>(number1) / number2;

    cout << "Quotient = " << quotient << endl;

    return 0;
}
```

```
Enter two numbers and I will divide
the first by the second for you.
82
33
Quotient = 2.48485

[Program finished]
```

```cpp

#include <iostream>
using namespace std;

int main()
{
    int num1 = 28;
    int num2 = 32;
    int num3 = 37;
    int num4 = 24;
    int num5 = 33;

    int sum;
    float average;

    sum = num1 + num2 + num3 + num4 +
num5;

    average = sum / 5.0;

    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}

```

```
Sum = 154
Average = 30.8

[Program finished]
```

```cpp

#include <iostream>
using namespace std;

int main() {
    int qty = 10;
    int salesReps = 3;
    double unitsEach;

    unitsEach = static_cast<double>(qty) / salesReps;

    cout << "Units Each = " << unitsEach << endl;

    return 0;
}

```

```
Units Each = 3.33333

[Program finished]
```

LAB NO. 04

Task 1

```
#include <iostream>
using namespace std;
int main () {
int number;
Cout << "Enter a number :";
cin >> number;
if (number > 0) {
Cout << " Number is positive ";
}
else{
cout << "Number is negative";
}
return 0;
}
```

output

Enter a number: 5
Number is positive

Enter a number: -4
Number is negative

Task#2

```
#include <iostream>
using namespace std;
int main(){
  int num;
Cout<<"Enter an integer:";
   Cin>>num;
 if(num>0){
   Cout<<"Number is positive";
 }
 else if(num<0){
    Cout<<"Number is negative";
 }
else{
  Cout<<"Number is zero";
  }
 return 0;
}
```

output:

Enter an integer: 8

Number is positive.

Task #03

```
#include <iostream>
using namespace std;
int main(){
  int mark;
Cout<<"Enter your marks:";
  cin>>mark;
if (mark>=75){
  Cout<<"Grade A";
}
else if (mark>=60){
  Cout<<"Grade B";
}
else if (mark >=45){
  Cout<<"Grade C";
}
else if (mark<30){
  Cout<<"Grade D";
}else{
  Cout<<"Grade not define";
}
  return 0;
}
```

Output

Enter your marks : 35
Grade is not defined

```
#include <iostream>
using namespace std;
int main(){
  int main;
Cout<< "Enter a number.";
  cin>>num;
if(num>0){
  Cout<<" Number is positive "<<endl;
if (num<10){
 Cout << "Number has 1 digit.";
}
else{
if (number<100){
Cout<< "Number has 2 digits.";
}
else{
if (num<1000){
   Cout<< "Number has 3 digit";
 }
  else{
   Cout<< "Number has 4 or more
    digits";
```

```
        }
      }
    }
  }
else{
  cout<<" Number is not positive ";
    }
    return 0;
  }
```

Output:-

Enter a number : 7

Number is positive

Number has 1 digit.

. —— .

```cpp

#include <iostream>
using namespace std;

int main() {
    int number;

    cout << "Enter a number: ";
    cin >> number;

    if (number > 0) {
        cout << "Number is positive";
    }
    else {
        cout << "Number is negative";
    }

    return 0;
}

```


Enter a number: 5
Number is positive
[Program finished]


Enter a number: -4
Number is negative
[Program finished]

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;

    cout << "Enter an integer: ";
    cin >> num;

    if (num > 0) {
        cout << "Number is positive";
    }
    else if (num < 0) {
        cout << "Number is negative";
    }
    else {
        cout << "Number is zero";
    }

    return 0;
}
```


Enter an integer: 8
Number is positive
[Program finished]

```cpp

#include <iostream>
using namespace std;

int main() {
    int mark;

    cout << "Enter your marks: ";
    cin >> mark;

    if (mark >= 75) {
        cout << "Grade A";
    }
    else if (mark >= 60) {
        cout << "Grade B";
    }
    else if (mark >= 45) {
        cout << "Grade C";
    }
    else if (mark < 30) {
        cout << "Grade D";
    }
    else {
        cout << "Grade not defined";
    }

    return 0;
}

```


Enter your marks: 35
Grade is not defined
[Program finished]

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;

    cout << "Enter a number: ";
    cin >> num;

    if (num > 0) {
        cout << "Number is positive" << endl;

        if (num < 10) {
            cout << "Number has 1 digit";
        }
        else {
            if (num < 100) {
                cout << "Number has 2 digits";
            }
            else {
                if (num < 1000) {
                    cout << "Number has 3 digits";
                }
                else {
                    cout << "Number has 4 or more digits";
                }
            }
        }
    }
    else {
        cout << "Number is not positive";
    }

    return 0;
```


Enter a number: 7
Number is positive
Number has 1 digit
[Program finished]